The quintic Ornstein-Uhlenbeck volatility model for joint SPX-VIX calibration

Speaker: Alice Magnanini (2198442)
Supervisor: Prof. Giorgia Callegaro

24/08/2026, Room 2BC30, 16:00

Abstract

The Volatility Index (VIX) has become, particularly since the financial crisis of 2008, one of the most widely used indices for trading and hedging the volatility in the US equity market, as measured by the S&P 500 Index (SPX). VIX is obtained by combining weighted prices of SPX puts and calls maturing in 30 days over a wide range of strikes: this makes the VIX a gauge of the market's expected volatility over the coming month. As a result, pricing VIX options for a given maturity T requires consistency with SPX options expiring up to $T + 30$ days. This motivates the importance of a joint modeling framework for SPX and VIX, since discrepancies between how the two are modeled can lead to mispricing and arbitrage opportunities.

The presentation will focus on the quintic Ornstein-Uhlenbeck (OU) volatility model, i.e., a stochastic volatility model where the volatility process is expressed as a fifth-degree polynomial function of a single Ornstein-Uhlenbeck process with suitable properties. What makes this model notable is its ability to jointly fit SPX-VIX smiles remarkably well using just 6 calibration parameters and an input curve that allows to match certain term structures. Because of its polynomial structure, the quintic OU volatility model is currently the simplest model able to achieve such good fits.

The goal of the presentation is to give the audience the background needed to understand the context and content of the paper, and then to explain the paper's results in depth. The talk will begin with the necessary technical preliminaries and an introduction to the joint SPX-VIX calibration problem. It will then describe the model itself, along with the pricing of VIX derivatives and some insight into pricing SPX derivatives within this framework. Finally, it will cover the calibration technique, the calibration results, and some insight into possible extensions of the model.

Main references

[1] Eduardo Abi Jaber, Camille Illand, Shaun Xiaoyuan Li. The quintic Ornstein-Uhlenbeck volatility model that jointly calibrates SPX & VIX smiles. Risk, 2023, Cutting edge section. ⟨hal-03909334v2⟩

[2] Shaun Xiaoyuan Li. The Quintic Volatility Model for SPX and VIX: A 360 Tour. Quantitative Finance [q-fin]. Université Paris 1 - Panthéon Sorbonne, 2024. English. ⟨NNT : ⟩. ⟨tel-04910511v2⟩

Contents

1 Introduction **3**

2 Preliminaries **3**
2.1 Standard & Poor's 500 Index (SPX) [Source: [Tra26a]] . . . 3
2.2 CBOE Volatility Index (VIX) [Source: [Cbo26a; Li24]] . . . 4
2.2.1 How the VIX is derived from SPX in the discrete case [Source: [Cbo25]] . . . 4
2.2.2 Understanding the VIX computation: an example [Source: [Cbo25]] . . . 4
2.2.3 Continuous time VIX formula [Source: [Neu94; Dem+99]] . . . 7
2.2.4 From the theoretical integral to the discrete VIX formula . . . 8
2.3 Implied volatility: skew and surface . . . 8
2.4 Stochastic volatility models . . . 9
2.5 Joint SPX & VIX calibration problem [Source: [GJR20]] . . . 9
2.5.1 What the quintic Ornstein-Uhlenbeck volatility model answers to [Source: [Li24]] . . . 9
2.6 The Ornstein-Uhlenbeck process . . . 10
2.6.1 Definition . . . 10
2.6.2 Basic properties . . . 10

3 The quintic Ornstein-Uhlenbeck volatility model **10**
3.1 Pricing VIX derivatives . . . 11
3.1.1 Towards the VIX formula used in [AIL23] . . . 11
3.1.2 An explicit expression for the VIX . . . 12
3.1.3 Pricing methodology for VIX options . . . 13
3.2 Pricing SPX derivatives . . . 13
3.3 SPX/VIX joint calibration . . . 13
3.3.1 Extracting the forward variance curve $\xi_0(\cdot)$ ([Source: [AIL23]]. . . . 14
3.3.2 Using a parametric forward variance curves when calibrating fewer slices to boost efficiency 15
3.3.3 Time-dependent H for longer maturities . . . 15
3.3.4 Author's recommendations on which model's extension to use for different cases . . . 15
3.4 Model calibration errors . . . 16

References **17**

1 Introduction

Before providing a detailed explanation of the SPX and VIX indices, we first present the following graphical visualization of their dynamics throughout April-December 2018. The figure highlights the fluctuations of both indices and, more importantly, should hint to the reader the need for a consistent joint modeling framework capable of capturing their interdependence.



Figure 1: Dynamics of SPX (above) and VIX (bottom) over the period April 2018 - December 2018. Source: *What is the VIX? A guide to the S&P 500 volatility index*, IG Bank.

Even without delving into the nature of the SPX and VIX indices, the figure suggests that it is natural to expect them to be used jointly, particularly for hedging purposes. Consider, for instance, the following two scenarios, provided in [Li24]:

- a trader from the sell-side pricing and hedging a product with a payoff that is exposed to both SPX & VIX indices,
- a portfolio manager from the buy-side constructing a portfolio and hedging his risk using SPX & VIX options.

In both cases, the knowledge of the covariance structure and joint density of SPX & VIX across time is beneficial and crucial. This motivates the importance of a joint modeling framework for SPX and VIX, since discrepancies between how the two are modeled can lead to mispricing and arbitrage opportunities.

2 Preliminaries

2.1 Standard & Poor's 500 Index (SPX) [Source: [Tra26a]]

Standard and Poor's 500 Index is a capitalization-weighted stock market index measuring the performance of 500 large publicly traded companies in the United States. This index covers a wide range of industries, including technology (e.g., Nvidia, Apple, Microsoft), healthcare (e.g., Johnson&Johnson, Pfizer), energy (e.g., ExxonMobil, Chevron), and finance (e.g., JPMorgan, Bank of America, Morgan Stanley); see [Tra26b] for the current complete list, and Table 1 which gives the market capitalization of the companies mentioned above.

SPX serves as a benchmark for the overall health of the U.S. stock market, as well as a reflection of the country's economic strength. Substantial gains in the S&P 500 are often interpreted as positive signals for the economy, while losses can be seen as indicators of potential trouble. Investors use this index as a barometer for their own portfolios and to assess the performance of individual stocks. The S&P 500 index plays a vital role in the financial world and is closely watched by analysts and policymakers.

Symbol	Company	Mkt cap. (USD)	Sector
NVDA	NVIDIA	5,2T	Technology
AAPL	Apple Inc.	4,51T	Technology
MSFT	Microsoft Corporation	3,59T	Technology
JNJ	Johnson & Johnson	651,25B	Healthcare
PFE	Pfizer Inc.	159,99B	Healthcare
XOM	Exxonmobil Holdings Corporation	678,92B	Energy
CVX	Chevron Corporation	405,57B	Energy
JPM	JPMorgan Chase & Co.	934,57B	Finance
BAC	Bank of America Corp	431,38B	Finance
MS	Morgan Stanley	336,42B	Finance

Table 1: Market capitalization in USD at date 22/08/2026 for some of the leader companies considered by SPX.

2.2 CBOE Volatility Index (VIX) [Source: [Cbo26a; Li24]]

The Volatility Index (VIX) has become, particularly since the financial crisis of 2008, one of the most widely used indices for trading and hedging the volatility in the US equity market, as measured by the SPX.

VIX is a financial index designed to be an up-to-the-minute market estimate of expected volatility of the SPX, and is calculated by using the midpoint of real-time SPX option bid/ask quotes. More specifically, the VIX Index is intended to provide an instantaneous measure of how much the market thinks the SPX will fluctuate in the 30 days from the time of each tick of the VIX Index.

2.2.1 How the VIX is derived from SPX in the discrete case [Source: [Cbo25]]

The generalized formula used in the VIX Index calculation is the following:

$$\frac{VIX^2}{100^2} = \sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T}\left[\frac{F}{K_0} - 1\right]^2 \tag{1}$$

Symbol	Description
VIX	Effective value of the VIX index;
σ	$\sigma = 100 \times VIX$, σ^2 represents the variance;
T	Time to expiration in years. Only SPX otions with more than 23 days and less than 37 days to the Friday SPX expiration are considered (source: [Cbo26a]);
F	Corresponding option-implied forward price: $F = K + e^{RT}(C - P)$, which derives directly from put-call parity: $C - P = e^{-RT}(F - K)$ (here, C is call price, P is put price, K is strike);
K_0	First strike equal to or otherwise immediately below the forward index level F;
K_i	Strike price of the i-th out-of-the-money option, which is a call if $K_i > K_0$, a put if $K_i < K_0$, both put and call if $K_i = K_0$;
ΔK_i	Half the gap between the neighbouring strikes: $\Delta K_i = (K_{i+1} - K_{i-1})/2$;
R	Risk-free interest rate to expiration;
$Q(K_i)$	Midpoint of the bid-ask spread for each option with strike K_i.

Table 2: Description of the terms appearing in the discrete VIX formula.

2.2.2 Understanding the VIX computation: an example [Source: [Cbo25]]

The goal of this note is to walk through, slowly and with commentary, the official Cboe example of a VIX computation provided in [Cbo25].

The idea is that VIX measures the market's expectation of 30-day volatility of the SPX, and it does so without assuming any option-pricing model. It is built purely from a portfolio of SPX put and call prices (which can be seen as a static replication of a variance swap). Because no maturity in the market is ever exactly 30 days, Cboe always uses two option expiries that bound from bottom and from above the 30-day mark (respectively, a *near-term* and a *next-term*) and interpolates between them.

Example setup. The calculation is performed on September 27, 2022, at 10:45:15 a.m. ET. Two option expiries are used:

- **Near-term**: SPX options expiring October 21, 2022 (24 calendar days away).
- **Next-term**: SPXW options expiring October 28, 2022 (31 calendar days away).

Step 1: computing the times to expiration. Everything is measured in minutes, then converted to a year fraction by dividing by the number of minutes in a year, $525600 = 365 \times 24 \times 60$.

$$T_1 = \frac{34484}{525600} = 0.0656088 \qquad T_2 = \frac{44954}{525600} = 0.0855289 \tag{2}$$

Here T_1 is the near-term time to expiry (about 24 days) and T_2 the next-term time to expiry (about 31 days). These two numbers alone already tell us: the true 30-day point sits between T_1 and T_2, which is exactly why we need to compute a variance for each term and then interpolate.

Step 2: computing the risk-free interests. Two separate risk-free rates are needed, one for each expiry, because interest rates depend on the horizon. Cboe interpolates the U.S. Treasury yield curve (via a cubic spline) to the exact expiration dates:

$$R_1 = 0.031664\% \quad \text{(near-term)}, \qquad R_2 = 0.028797\% \quad \text{(next-term)}. \tag{3}$$

These rates will appear only inside a discounting factor e^{RT} later on and their effect on the final answer is small, but they must be included for the calculation to be arbitrage-consistent.

Step 3: computing the forward price F and the reference strike K_0. We need to compute the forward price because VIX is meant to measure volatility around the forward level of the index, not around today's spot price. Rather than trust a possibly stale spot quote, Cboe extracts the forward price directly from option prices, using put-call parity. For call price C and put price P at the same strike K, put-call parity gives $C - P = e^{-RT}(F - K)$, which rearranges to

$$F = K + e^{RT}(C - P) \tag{4}$$

This is applied at the strike where C and P are closest to each other (the strike is then called the at-the-money strike), since that is where the parity relationship is least sensitive to bid-ask noise.

We now plot the data provided in order to deduce the at-the-money strikes. Call and put prices are computed as midpoints of bid/ask prices. The at-the-money strike is the multiple of 5 that produces the difference nearest to 0. We deduce that the at-the-money strike is 1965 for the near-term and 1960 for the next-term. The forward prices are then computed taking call and put prices corresponding to the at-the-money strikes:

$$F_1 = 1965 + e^{(0.00031664\%)(0.0656088)} \times (21.05 - 23.15) = 1962.89996 \qquad \text{(near-term)} \tag{5}$$

$$F_2 = 1960 + e^{(0.00028797\%)(0.0855289)} \times (27.30 - 24.90) = 1962.40006 \qquad \text{(next-term)} \tag{6}$$

K_0 is simply defined as the first listed strike at or below F. Since $F_1 = 1962.9$ and $F_2 = 1962.4$, and strikes are quoted in multiples of 5 in this example, we get

$$K_{0,1} = 1960 \qquad K_{0,2} = 1960. \tag{7}$$

K_0 matters because it is the pivot point of the whole calculation: every strike below K_0 will be treated as a put, every strike above K_0 as a call (see next step).

Step 4: selecting which options to use. For each term, build one ladder of strikes:

- all **out-of-the-money puts**, i.e. strikes below K_0, walking downward and stopping as soon as two consecutive strikes are found with a zero bid (illiquid, unusable quotes);
- all **out-of-the-money calls**, i.e. strikes above K_0, walking upward with the same stopping rule;
- **both** the put and the call exactly at K_0 (their prices are averaged into a single number).

The reason behind this choice of options relies on Carr-Madan / Breeden-Litzenberger decomposition; see Section 2.2.3 for details. Returning back to the example, for the near-term the $K_0 = 1960$ contribution is $(24.25 + 21.30)/2 = 22.775$; for the next-term, $(27.30 + 24.90)/2 = 26.10$. Every option price used from here on is the midpoint of the bid and the ask (never the last traded price) since the bid-ask midpoint is the standard proxy for "fair value" when no model is assumed.

Step 5: assembly the VIX. This is the heart of the calculation. For each term separately (drop the subscript 1 and 2 for readability), the variance is given by Eq. (1). The intuition is that the sum $\sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i)$ is a

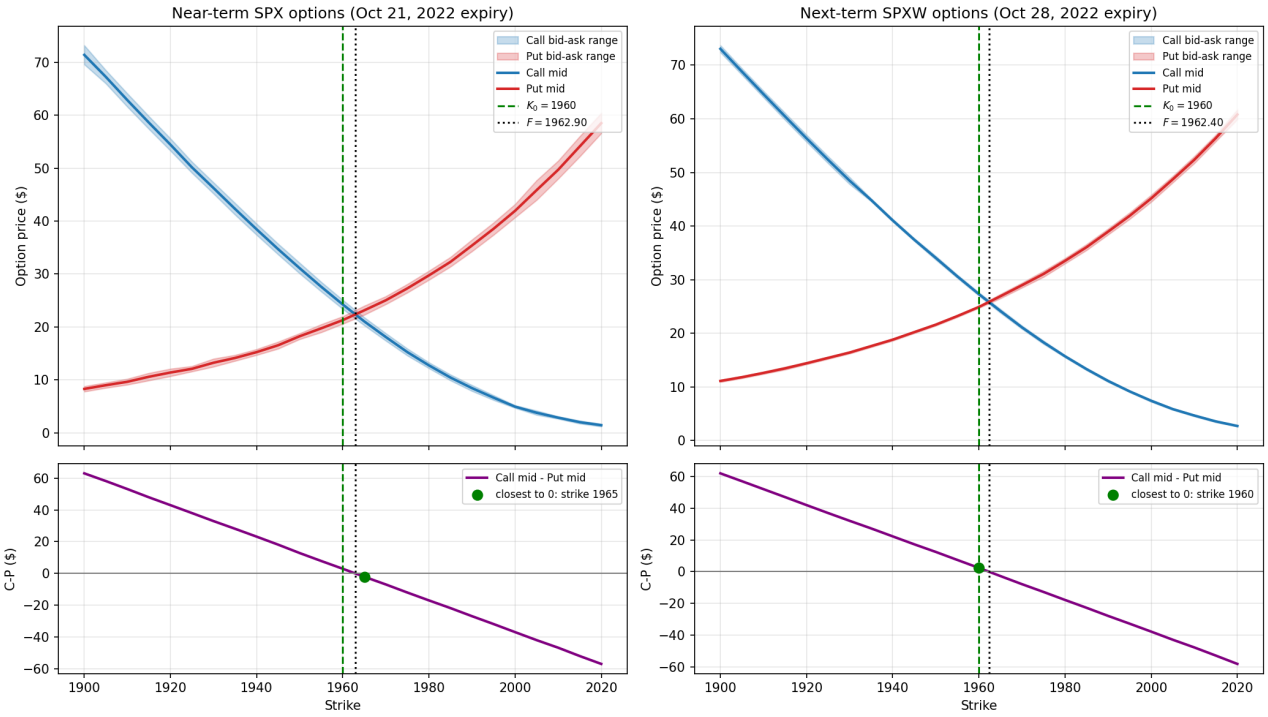


Figure 2: Data-plotting of call-put prices that allows to find the at-the-money strike, i.e., the strike at which call and put prices are the same. Note that only strikes that are multiples of 5 are considered in this example. Both plots show also their own forward price F and the corresponding nearest strike K_0.

discretized integral over all out-of-the-money option prices. This is the discrete version of a classical continuous replication formula for a log-contract:

$$\mathbb{E}\Big[\int_0^T \sigma_u^2\, du\Big] \;\propto\; \int_0^\infty \frac{2}{K^2}\,\text{price}(K)\, dK. \tag{8}$$

In words: every single option in the strike ladder contributes a little sliver of "expected variance", weighted by how far out-of-the-money it is ($1/K_i^2$) and by the width of strike-space it represents (ΔK_i). The correction term $\frac{1}{T}\big[F/K_0 - 1\big]^2$ is a small adjustment that accounts for using K_0 rather than the (unobservable, continuous) forward price F itself as the split point between puts and calls. See Sections 2.2.3 and 2.2.4 for further details.

Plugging in the numbers, for the near-term (24-day) options, summing every contribution in the strike ladder gives $\sum_i \frac{\Delta K_i}{K_i^2} e^{R_1 T_1} Q(K_i) = 0.0006320516$, so that:

$$\frac{2}{T_1}\sum_i \frac{\Delta K_i}{K_i^2}\, e^{R_1 T_1} Q(K_i) = 0.019267, \qquad \frac{1}{T_1}\left[\frac{F_1}{K_{0,1}} - 1\right]^2 = \frac{1}{0.0656088}\left[\frac{1962.89996}{1960} - 1\right]^2 = 0.00003337 \tag{9}$$

$$\Rightarrow \quad \sigma_1^2 = 0.019267 - 0.00003337 = 0.019233906 \tag{10}$$

For the next-term (31-day) options, the analogous computation gives:

$$\frac{2}{T_2}\sum_i \frac{\Delta K_i}{K_i^2}\, e^{R_2 T_2} Q(K_i) = 0.019441, \qquad \frac{1}{T_2}\left[\frac{F_2}{K_{0,2}} - 1\right]^2 = 0.00001753 \tag{11}$$

$$\Rightarrow \quad \sigma_2^2 = 0.019441 - 0.00001753 = 0.019423884 \tag{12}$$

Example of computation of a single contribution. Take the near-term 1370 put, the lowest strike included in the ladder. Its midpoint price is $Q = 0.2$, and since it sits at the edge of the ladder, $\Delta K = 1375 - 1370 = 5$. Its contribution to the sum is:

$$\frac{\Delta K_{1370}}{K_{1370}^2}\, e^{R_1 T_1}\, Q(1370) = \frac{5}{1370^2}\, e^{0.00031664\% \times 0.0656088} \times 0.2 = 0.0000005328. \tag{13}$$

Every one of the roughly 80 near-term and 90 next-term strikes contributes one such small number; adding them all up (and multiplying by $2/T$) is what produces 0.019267 and 0.019441 above.

Step 6: Interpolating to a constant 30-day maturity. We now have two variances, σ_1^2 (at $T_1 \approx 24$ days) and σ_2^2 (at $T_2 \approx 31$ days). VIX needs *exactly* 30 days, so a time-weighted average is taken, using minute-counts: $N_{30} = 43200$ minutes (30 days), $N_{T_1} = 34484$, $N_{T_2} = 44954$, $N_{365} = 525600$.

$$VIX = 100 \times \sqrt{\left\{ T_1\, \sigma_1^2 \left[\frac{N_{T_2} - N_{30}}{N_{T_2} - N_{T_1}} \right] + T_2\, \sigma_2^2 \left[\frac{N_{30} - N_{T_1}}{N_{T_2} - N_{T_1}} \right] \right\} \times \frac{N_{365}}{N_{30}}} \tag{14}$$

The two bracketed fractions are simply linear-interpolation weights: since 30 days sits between 24 and 31 days, the near-term variance gets weight $\frac{44{,}954-43{,}200}{44{,}954-34{,}484} \approx 0.168$ and the next-term variance gets the complementary weight ≈ 0.832. The final $\times N_{365}/N_{30}$ rescales the result to an annualized, 30-day-horizon variance, and the outer square root converts variance into volatility. Multiplying by 100 turns a decimal (like 0.14) into the familiar VIX quoting convention (like 14).

Step 7: final number. Plugging in $\sigma_1^2 = 0.019233906$ and $\sigma_2^2 = 0.019423884$:

$$VIX = 100 \times \sqrt{0.0656088 \times 0.019233906 \times 0.168 + 0.0855289 \times 0.019423884 \times 0.832} \times \frac{525600}{43200} \tag{15}$$

$$\Rightarrow VIX = 100 \times 0.13927842 = 13.93. \tag{16}$$

Summary of the logical flow:

[1] Pick two option expiries bracketing 30 days (near-term, next-term);

[2,3] For each expiry (near-term, next-term): get the risk-free rate, extract the forward price F from put-call parity, find K_0 just below F;

[4,5] For each expiry: build a strike ladder of OTM puts/calls around K_0, and sum their midpoint prices weighted by $\Delta K_i / K_i^2$;

[6,7] Linearly interpolate the two variances to a constant 30-day horizon, annualize, take the square root, and multiply by 100.

Note that *VIX* is called *model-free*: nowhere in this derivation did we assume a volatility model (no Black-Scholes, no Heston); we only assumed the absence of arbitrage (put-call parity) and a discretized replication of a variance swap from observed option prices.

2.2.3 Continuous time VIX formula [Source: [Neu94; Dem+99]]

The starting point is a variance swap: a contract paying the realized variance of log returns over $[0, T]$. The key result (from classical references [Neu94; Dem+99]) is that under continuous monitoring and a diffusion (no-jump) assumption, expected realized variance can be written as the risk-neutral expectation of a log-payoff:

$$\sigma^2 := \mathbb{E}\left[\int_0^T \sigma_t^2\, dt \right] = \frac{2}{T}\, \mathbb{E}\left[\int_0^T \frac{dS_t}{S_t} - \ln \frac{S_T}{S_0} \right]. \tag{17}$$

The first term (the sum of returns) can be replicated by a simple dynamically-rebalanced futures/forward position. The hard part is the $-\ln(S_T/S_0)$ piece (the so called "log-contract") because a log payoff cannot be bought directly. However, any smooth payoff can be spanned by a continuum of European options via the Carr-Madan / Breeden-Litzenberger decomposition [Bos21]:

$$f(S_T) = f(\kappa) + f'(\kappa)(S_T - \kappa) + \int_0^\kappa f''(K)(K - S_T)^+\, dK + \int_\kappa^\infty f''(K)(S_T - K)^+\, dK, \tag{18}$$

valid for any reference point κ. Applying this to $f(x) = -\ln(x/\kappa)$, whose second derivative is $f''(x) = 1/x^2$, gives the density needed to statically replicate a log payoff out of puts and calls (and from this follows the $1/K^2$ weighting in the discrete VIX formula). Choosing $\kappa = F$ (the forward price) makes the linear term vanish in expectation (since forwards are $\mathbb{Q}$-martingales, so $F = \mathbb{E}[S_T]$). The theoretical continuous-strike variance formula is then exact (since the integral is measurable):

$$\sigma^2 = \frac{2}{T}\left[\int_0^F \frac{e^{RT}}{K^2} P(K)\, dK + \int_F^\infty \frac{e^{RT}}{K^2} C(K)\, dK \right] = \frac{2}{T} \int_0^\infty \frac{e^{RT}}{K^2} Q(K)\, dK, \tag{19}$$

with no extra correction term. Note that $P(K), C(K)$ express respectively the prices of put and call options with strike K with maturity T. This explains the choice of the options in Step 4 of Section 2.2.2, i.e., ideally, puts with strike less than F and calls with strike greater than F. Being the type of the option uniquely determined

by the strike, we can put together the integrals and express the price of the options as a function: $Q(K)$. The correction term that appears in (1) derives from the fact that real markets do not offer a strike exactly at the forward price, so we have to substitute the theoretical, unobservable F with the realistic strike K_0, whose rules of choice are suitable for applying a Taylor expansion (see Section 2.2.4 for details).

2.2.4 From the theoretical integral to the discrete VIX formula

Two approximations turn this continuum integral (19) into the practical Cboe formula (1):

1. **Using K_0 instead of F as the expansion point.** The clean cancellation above requires $\kappa = F$ exactly, but F is generally not a traded strike; only K_0, the nearest listed strike below F, is available. Expanding around $\kappa = K_0$ instead of F leaves a residual (approximately quadratic) term: indeed $f(K_0) = 0$, $f'(K_0) = -1/K_0$, so applying Carr-Madan / Breeden-Litzenberger decomposition we get:

$$-\ln\frac{S_T}{K_0} = -\frac{1}{K_0}(S_T - K_0) + \int_0^{K_0} \cdots + \int_{K_0}^{\infty} \cdots =: -\frac{1}{K_0}(S_T - K_0) + I \tag{20}$$

   Rewrite the LHS as $-\ln S_T/K_0 = -\ln S_T/F - \ln F/K_0$ and take expectations; recalling that I and $-\ln F/K_0$ are measurable and $F = \mathbb{E}[S_t]$, then

$$\mathbb{E}\left[-\ln\frac{S_T}{F}\right] = \ln\frac{F}{K_0} - \frac{1}{K_0}(F - K_0) + I \approx -\frac{1}{2}\left[\frac{F}{K_0} - 1\right]^2 + I \tag{21}$$

   where we have used Taylor's expansion $\ln(1+x) = x - x^2/2 + O(x^3)$ for $x = F/K_0 - 1$, which is $\gtrsim 0$ by definition of K_0. Hence the variance becomes:

$$\sigma^2 = \frac{2}{T}\mathbb{E}\left[-\ln\frac{S_T}{F}\right] = \frac{2}{T}I - \frac{1}{T}\left[\frac{F}{K_0} - 1\right]^2. \tag{22}$$

2. **Discretization of the strike integral.** Only listed strikes K_i with spacing ΔK_i exist, so

$$I = \int_0^{\infty} \frac{e^{RT}}{K^2}\, Q(K)\, dK \;\longrightarrow\; \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i), \tag{23}$$

   using mid-quote prices $Q(K_i)$.

Combining both approximations reproduces the Cboe VIX formula (1) exactly.

2.3 Implied volatility: skew and surface

An option's price is a function of several observable inputs (spot/forward, strike, maturity, interest rate) and one unobservable input: the volatility of the underlying. Under the Black-Scholes model, the price of a European call is

$$C_0^{BS}(K, T; \sigma) = F(T)\,\mathcal{N}(d_1) - K\,\mathcal{N}(d_2), \qquad d_1 = \frac{\log(F(T)/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \tag{24}$$

where $\mathcal{N}(\cdot)$ is the standard normal cumulative density function of the standard Gaussian distribution and $F(T)$ is the forward price of the underlying for maturity T.

Given a market-observed price $C_0(K,T)$, *implied volatility* $\sigma(K,T)$ is defined as the unique value of σ that makes the Black-Scholes formula reproduce that price:

$$C_0(K,T) = C_0^{BS}\big(K, T; \sigma(K,T)\big). \tag{25}$$

Existence and uniqueness follow because C_0^{BS} is strictly increasing in σ (the so-called "positive *vega*"), so the inversion is well posed and can be solved numerically (e.g., bisection).

If the Black-Scholes assumptions held exactly (constant σ, no jumps, continuous trading, lognormal returns), $\sigma(K,T)$ would be the same number for every K and T, equal to the constant instantaneous volatility of the underlying. In real markets this is never observed: the map $K \mapsto \sigma(K,T)$ is not flat. This curve is the *implied volatility smile* (or *skew*) for maturity T, and the collection of smiles across maturities is the *implied volatility surface* $(K,T) \mapsto \sigma(K,T)$. Note that implied volatility skew can be equivalently expressed as function of the log-moneyness $\log(K/S_0)$. Figure 3 shows typical SPX and VIX volatility skews.

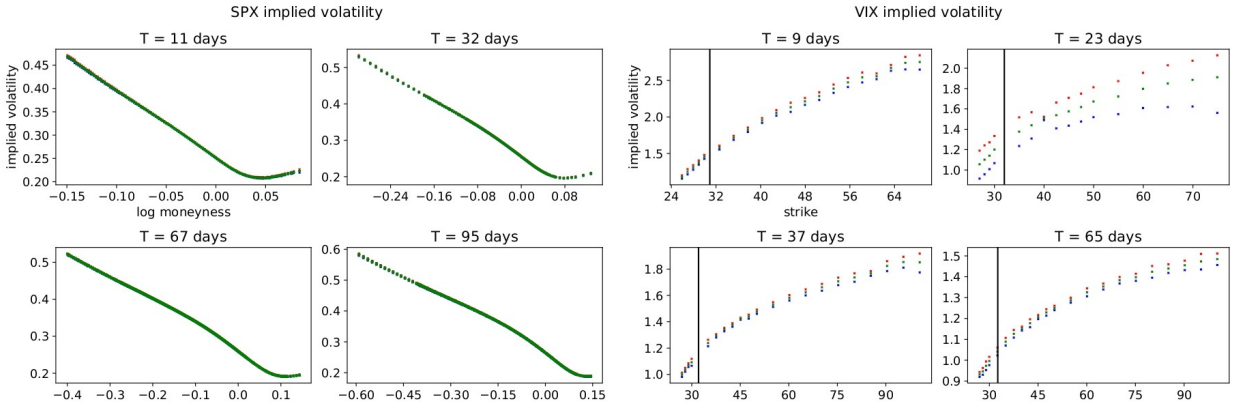


Figure 3: SPX and VIX smiles (bid/ask in blue/red, mid-price in green) and VIX futures (vertical black lines) on 13 July 2020. [Source: [Li24].]

2.4 Stochastic volatility models

Real market's implied volatility skews show that a single Black–Scholes volatility σ can't reproduce market option prices across strikes: indeed, they show that the market's risk-neutral distribution isn't lognormal. The natural fix is to stop treating σ as a constant and instead let it be random itself:

$$\frac{dS_t}{S_t} = \sigma_t dB_t, \qquad \sigma_t = f(\text{some auxiliary driving process}), \tag{26}$$

with the driving process somehow correlated to the stock's own Brownian motion. This is the *general stochastic volatility (SV) framework*: Heston, SABR, 3/2, and our quintic OU volatility models are all instances.

2.5 Joint SPX & VIX calibration problem [Source: [GJR20]]

Since 2004, investors have been able to trade VIX futures: they provide market participants with a variety of opportunities to implement their view using volatility trading strategies, including risk management and portfolio diversification ([Source: [Cbo26b]]). Subsequently in 2006, CBOE introduced VIX options providing market participants with another tool to manage volatility. VIX options enable market participants to hedge portfolio volatility risk distinct from market price risk and trade based on their view of the future direction or movement of volatility [Source: [GJR20]]. Those products are now among the most liquid financial instruments in the world. There are indeed more than 500,000 VIX options traded each day, with most of the liquidity concentrated on the first three monthly contracts. Nevertheless, despite that more vega is now traded in the VIX market than in the SPX market, the wide bid-ask spreads in the VIX options market betray its lack of maturity. One of the reasons underlying these wide spreads is that *the market lacks a reliable pricing methodology for VIX options*; since the VIX is by definition a derivative of the SPX, any reasonable methodology must necessarily be consistent with the pricing of SPX options.

Designing a model that jointly calibrates SPX and VIX options prices is known to be extremely challenging. Indeed, this problem is sometimes considered to be the *holy grail of volatility modeling*. We will simply refer to it as the *joint SPX&VIX calibration problem*.

Figure 3 graphically shows the challenges that the joint calibration problem faces:

1. a multitude of instrumentsto be calibrated across several maturities,
2. a steep upward-sloping VIX volatility smile,
3. the critical at-the-money (ATM) SPX skew, which becomes more pronounced for smaller maturities.

2.5.1 What the quintic Ornstein-Uhlenbeck volatility model answers to [Source: [Li24]]

Given the importance of consistent modeling of SPX & VIX, we want to find a *consistent* model that can:

1. calibrate across several maturities to European call and put options on SPX & VIX together with VIX futures, i.e. a total of three instruments to be calibrated across various maturities. Calibrating to the

VIX future is just as important as to the VIX option price, as the VIX future price is (i) a tradable asset listed on the CBOE, and (ii) the input for the computation of implied volatility of VIX options via the Black–Scholes formula. A wrong VIX future price would translate into wrong VIX option prices at the moneyness level;

2. fit SPX smiles with maturities up to one-month ahead of the largest VIX calibrated maturities for consistency of the two indices since VIX is forward-looking and reflects the expected volatility of SPX for the next 30 days.

2.6 The Ornstein-Uhlenbeck process

2.6.1 Definition

Let W be a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{Q})$. The (time-homogeneous) Ornstein-Uhlenbeck process X is the unique strong solution of the linear stochastic differential equation

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t, \qquad X_0 = x_0, \tag{27}$$

where $\kappa > 0$ is the *mean-reversion speed*, $\theta \in \mathbb{R}$ is the *long-run mean* (the level X reverts towards), and $\sigma > 0$ is the *volatility* (or *vol-of-vol*, when X itself drives another process's volatility, as we will see in the quintic OU volatility model). The drift $\kappa(\theta - X_t)$ pulls X back towards θ at a rate proportional to its current distance from θ; the diffusion term $\sigma\,dW_t$ continuously perturbs it.

2.6.2 Basic properties

1. Explicit strong solution. Applying Itô's formula to $e^{\kappa t}X_t$ gives the closed-form (stochastic convolution) representation

$$X_t = \theta + (x_0 - \theta)e^{-\kappa t} + \eta \int_0^t e^{-\kappa(t-s)}\,dW_s. \tag{28}$$

   With $\theta = 0$ (the special case in [AIL23]), this simplifies to $X_t = x_0 e^{-\kappa t} + \eta \int_0^t e^{-\kappa(t-s)}\,dW_s$.

2. **Gaussianity.** Because X is defined by a linear SDE driven by Brownian motion, X_t is a Gaussian random variable for every t, and in fact $(X_{t_1}, \ldots, X_{t_n})$ is jointly Gaussian for any finite collection of times: X is a Gaussian process.

3. **Mean and variance.** From the explicit solution, taking expectations and using the Itô isometry:

$$\mathbb{E}[X_t] = \theta + (x_0 - \theta)e^{-\kappa t}, \qquad \mathrm{Var}(X_t) = \eta^2 \int_0^t e^{-2\kappa(t-s)}\,ds = \frac{\eta^2}{2\kappa}\Big(1 - e^{-2\kappa t}\Big). \tag{29}$$

   Note $\mathrm{Var}(X_t)$ does not depend on θ: only the mean is shifted by the long-run level, while the spread around it is governed by κ, η alone. With $x_0 = \theta = 0$, $\mathbb{E}[X_t] = 0$ for all t, and

$$\mathrm{Var}(X_t) = \frac{\varepsilon^{2H}}{1-2H}\Big(1 - e^{-(1-2H)\varepsilon^{-1}t}\Big), \tag{30}$$

   using $\eta^2/(2\kappa) = \varepsilon^{2H-1}/\big(2(1/2-H)\varepsilon^{-1}\big) = \varepsilon^{2H}/(1-2H)$.

4. **Markov property.** X is a (time-homogeneous) Markov process: conditional on $\mathcal{F}_T$, the future path $(X_u)_{u\geq T}$ depends on the past only through the current value X_T.

3 The quintic Ornstein-Uhlenbeck volatility model

This model was developed to answer the following questions [Source: [Li24]]:

1. *Can joint calibration of SPX & VIX volatility smiles be achieved without appealing to multiple-factors, jumps, roughness or path-dependency?*

2. *Is joint calibration possible with parsimonious one-factor continuous Markovian models?*

3. *If so, can we do it in a tractable way?*

Consider a two-dimensional Brownian motion $(W, W^\perp)$ on a risk-neutral filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{Q})$.

Stochastic volatility framework. The stock price is driftless under $\mathbb{Q}$ (no rates/dividends), as in a generic stochastic volatility model:

$$\frac{dS_t}{S_t} = \sigma_t \, dB_t, \qquad B = \rho W + \sqrt{1-\rho^2}\, W^\perp, \tag{31}$$

with $\rho \in [-1, 1]$ the correlation between the stock's driver B and the volatility's driver W.

Volatility modeling. Volatility is modeled as a *degree-5 polynomial* of a single driving process X:

$$\sigma_t = \sqrt{\xi_0(t)}\, \frac{p(X_t)}{\sqrt{\mathbb{E}[p(X_t)^2]}}, \qquad p(x) = \alpha_0 + \alpha_1 x + \alpha_3 x^3 + \alpha_5 x^5, \tag{32}$$

where $\alpha_i \geq 0$ and $\xi_0 \in L^2([0,T], \mathbb{R}_+)$ for any $T > 0$. Why these design choices? [Source: [AIL23]]

- The normalization $\sqrt{\mathbb{E}[p(X_t)^2]}$ allows the input curve $\xi_0(\cdot)$ to carry alone the market's forward-variance term structure (matching $\mathbb{E}[\int_0^t \sigma_s^2 \, ds] = \int_0^t \xi_0(s) \, ds$), freeing X_t to be a pure zero-mean fluctuation (i.e., $\theta = 0$);
- Taking p a polynomial of degree five allows us to reproduce the upward slope of the VIX smile; restricting the coefficients α to be non-negative (with $\alpha_2 = \alpha_4 = 0$) ensures the sign of the at-the- money skew to be the same as ρ, see [AIL22] for more details, as well as ensuring the martingale property of S, whenever $\rho \leq 0$ and $\alpha_5 \geq 0$, see Appendix A of [AIL23].

The driving process of volatility. We take a zero-mean Ornstein-Uhlenbeck process X as the driving process for the volatility:

$$dX_t = -(1/2 - H)\varepsilon^{-1} X_t \, dt + \varepsilon^{H-1/2} \, dW_t. \tag{33}$$

with $H \in (-\infty, \frac{1}{2})$, $\epsilon > 0$ (sufficiently small). Note that:

- X has *fast mean-reversion*: indeed, the mean-reversion speed is $\kappa = (1/2 - H)\varepsilon^{-1}$, which is large if $\frac{1}{2} - H > 0$ and ε is small;
- X has *large vol-of-vol*: indeed, $\sigma = \varepsilon^{H-1/2} = (1/\varepsilon)^{\frac{1}{2}-H}$, which is large if $\frac{1}{2} - H > 0$ and $1/\varepsilon$ is large, i.e., if ε is small.

The role of $H \in (-\infty, \frac{1}{2})$ is crucial [Source: [AIL23]]: it makes the model more flexible; it makes it achieve better fits (as a matter of fact, it generalizes a broad class of models: jump models for $H \leq \frac{1}{2}$, Foque's fast regimes for $H = 0$, rough volatility models for $H \in (0, \frac{1}{2})$, which plays the role of the Hurst index); and finally introduces time-stabilization into the model.

Note that the model will not calibrate κ, σ; rather, it calibrates ε, H and then expresses κ, σ as functions of ε, H. This method allows to achieve better stabilization through time.

Therefore: *the quintic Ornstein-Uhlenbeck volatility model consists simply in the equations* (31), (32) *and* (33). *Fixing* $\varepsilon = 1/52$ *leaves just six calibratable numbers,* $\Theta = \{\alpha_0, \alpha_1, \alpha_3, \alpha_5, \rho, H\}$, *plus the curve* $\xi_0(\cdot)$.

A note on the number of parameters [Source: [AIL23]]: numerical experiments show no significant adverse impact on the joint calibration quality by narrowing the number of parameters.

3.1 Pricing VIX derivatives

3.1.1 Towards the VIX formula used in [AIL23]

Work under risk-neutral dynamics with zero interest and dividends: $\frac{dS_t}{S_t} = \sigma_t \, dB_t$. Assume $\rho \leq 0$; then, S is a $\mathbb{Q}$-martingale (see Appendix A of [AIL23]). By definition, the VIX at a future date T is the annualized risk-neutral expected variance of log-returns of S over the next $\Delta = 30$ days, conditional on $\mathcal{F}_T$:

$$VIX_T^2 = \frac{100^2}{\Delta}\, \mathbb{E}\left[\int_T^{T+\Delta} \sigma_u^2 \, du \,\middle|\, \mathcal{F}_T \right]. \tag{34}$$

Since S has no drift, $d(\log S_t) = -\frac{1}{2}\sigma_t^2 \, dt + \sigma_t \, dB_t$. Integrating from T to $T + \Delta$:

$$\log(S_{T+\Delta}/S_T) = -\frac{1}{2} \int_T^{T+\Delta} \sigma_u^2 \, du + \int_T^{T+\Delta} \sigma_u \, dB_u. \tag{35}$$

Rearranging we recover the *log-contract identity*:

$$\int_T^{T+\Delta} \sigma_u^2 \, du = -2 \log(S_{T+\Delta}/S_T) + 2 \int_T^{T+\Delta} \sigma_u \, dB_u. \tag{36}$$

If $\int_T^{T+\Delta} \sigma_u \, dB_u$ is a true martingale (not merely a local one), its conditional expectation vanishes, so

$$\mathbb{E}\left[\int_T^{T+\Delta} \sigma_u^2 \, du \,\middle|\, \mathcal{F}_T \right] = -2 \, \mathbb{E}[\log(S_{T+\Delta}/S_T) \,|\, \mathcal{F}_T] \, . \tag{37}$$

Set $\xi_T(u) := \mathbb{E}[\sigma_u^2 \mid \mathcal{F}_T]$ for $u \geq T$ (the forward instantaneous variance seen from T). By Fubini's theorem,

$$\mathbb{E}\left[\int_T^{T+\Delta} \sigma_u^2 \, du \,\middle|\, \mathcal{F}_T \right] = \int_T^{T+\Delta} \xi_T(u) \, du. \tag{38}$$

Hence we recover the formula for the VIX used in [AIL23]:

$$VIX_T^2 = -\frac{2}{\Delta} \, \mathbb{E}[\log(S_{T+\Delta}/S_T) | \mathcal{F}_T] \times 100^2 = \frac{100^2}{\Delta} \int_T^{T+\Delta} \xi_T(u) \, du \tag{39}$$

which is exactly equation (3.1) of the paper. It states two equivalent things: VIX_T^2 is proportional to (i) minus the expected log return over the next 30 days, annualized, or (ii) the average forward variance over the window $[T, T+\Delta]$.

3.1.2 An explicit expression for the VIX

Fix $u \geq T$ and rewrite X as follows:

$$X_u = X_T \, e^{-(1/2-H)\varepsilon^{-1}(u-T)} + \varepsilon^{H-1/2} \int_T^u e^{-(1/2-H)\varepsilon^{-1}(u-s)} \, dW_s =: Z_T^u + G_T^u, \tag{40}$$

where Z_T^u is $\mathcal{F}_T$-measurable (a deterministic function of X_T) and G_T^u is Gaussian, independent of $\mathcal{F}_T$, with

$$G_T^u \sim \mathcal{N}\left(0, \; \frac{\varepsilon^{2H}}{1-2H}\left(1 - e^{-(1-2H)\varepsilon^{-1}(u-T)}\right)\right), \tag{41}$$

which comes from the well-know relation $\int_t^T f(s) dW_s \sim \mathcal{N}\Big(0, \int_t^T f(s)^2 ds\Big)$.

Set $g(u) := \mathbb{E}[p(X_u)^2]$. Squaring p and collecting terms via the discrete convolution $(\alpha * \alpha)_k := \sum_{j=0}^k \alpha_j \alpha_{k-j}$,

$$p(X_u)^2 = \left(\sum_{k=0}^5 \alpha_k X_u^k \right)^2 = \sum_{k=0}^{10} (\alpha * \alpha)_k \, X_u^k, \tag{42}$$

so that

$$\xi_T(u) = \mathbb{E}\big[\sigma_u^2 \mid \mathcal{F}_T\big] = \frac{\xi_0(u)}{g(u)} \, \mathbb{E}\left[\sum_{k=0}^{10} (\alpha * \alpha)_k \, X_u^k \,\middle|\, \mathcal{F}_T \right]. \tag{43}$$

Since Z_T^u is $\mathcal{F}_T$-measurable and G_T^u is independent of $\mathcal{F}_T$, substituting $X_u = Z_T^u + G_T^u$ into X_u^k and expanding binomially:

$$\xi_T(u) = \frac{\xi_0(u)}{g(u)} \sum_{k=0}^{10} \sum_{i=0}^{k} (\alpha * \alpha)_k \binom{k}{i} \left(X_T \, e^{-(1/2-H)\varepsilon^{-1}(u-T)} \right)^i \mathbb{E}\big[(G_T^u)^{k-i}\big], \tag{44}$$

with $\binom{k}{i} = k!/((k-i)!\, i!)$. All the randomness on the right that is not known given $\mathcal{F}_T$ has now been isolated into the factor $\mathbb{E}[(G_T^u)^{k-i}]$, which is a deterministic number (Gaussian moments of a known variance). Indeed, for $Y \sim \mathcal{N}(0, \sigma_Y^2)$,

$$\mathbb{E}[Y^p] = \begin{cases} 0 & p \text{ odd} \\ \sigma_Y^p \, (p-1)!! & p \text{ even}, \end{cases} \tag{45}$$

with $(p-1)!!$ the double factorial. Every term $\mathbb{E}[(G_T^u)^{k-i}]$ is therefore available in closed form, with no numerical integration over the volatility process required.

Hence, the VIX is polynomial in X_T: substituting (44) into (39) and swapping the (finite) sums with the integral over u, every power of X_T collects into a single coefficient:

$$\mathrm{VIX}_T^2 = \frac{100^2}{\Delta} \sum_{i=0}^{10} \beta_i \, X_T^i =: h(X_T) \tag{46}$$

where

$$\beta_i = \sum_{k=i}^{10} (\alpha * \alpha)_k \binom{k}{i} \int_T^{T+\Delta} \frac{\xi_0(u)}{g(u)} \, \mathbb{E}\big[(G_T^u)^{k-i}\big] \, \Big(e^{-(1/2-H)\varepsilon^{-1}(u-T)}\Big)^i \, du. \tag{47}$$

The integral inside each β_i depends only on T (not on X_T) and can be computed, at least numerically, for essentially any choice of $\xi_0(\cdot)$. Equation (46) is the central tractability result of the model: VIX_T^2, an object that in general stochastic volatility models requires simulation or a PDE to characterize, is here an explicit polynomial of degree 10 in the single variable X_T.

3.1.3 Pricing methodology for VIX options

Since X_T is Gaussian with mean 0 and variance $\sigma_{X_T}^2 = \frac{\varepsilon^{2H}}{1-2H}(1 - e^{-(1-2H)\varepsilon^{-1}T})$, pricing VIX derivatives with payoff function Φ is immediate by integrating directly against the standard Gaussian density:

$$\mathbb{E}\big[\Phi(\mathrm{VIX}_T)\big] = \mathbb{E}\Big[\Phi\big(\sqrt{h(X_T)}\big)\Big] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \Phi\Big(\sqrt{h(\sigma_{X_T} x)}\Big) \, e^{-x^2/2} \, dx. \tag{48}$$

Recall our discussion in Section 2.5 regarding the lack of a pricing methodology for VIX options: in this context, the quintic OU model provides the most satisfactory answer, as it yields a theoretical result that is both mathematically tractable and straightforward to implement numerically.

3.2 Pricing SPX derivatives

The pricing of SPX derivatives, unlike VIX derivatives, do not admit a closed-form expression: their payoff depends on the full path of the stock price rather than on the volatility factor at a single instant, so we resort to Monte Carlo simulation. To keep this simulation both accurate and efficient, two techniques are combined.

- First, since the volatility-driving process X is an Ornstein-Uhlenbeck process, it admits an exact simulation scheme with no time-discretization bias: this could possibly represent a critical feature given the model's fast-mean-reversion, large-vol-of-vol regime, which would require prohibitively small time steps under a naive Euler discretization.
- Second, for the stock price itself, which must still be simulated via an Euler scheme, we apply the "turbocharging" method in [MP18]: conditional on the volatility path, the component of the stock price driven by the independent noise is lognormally distributed, admitting a quasi-closed-form price; this structure is exploited as a control variate, together with antithetic sampling, to substantially reduce Monte Carlo variance.

Together, these techniques allow SPX derivatives to be priced accurately and at low computational cost despite the absence of a closed-form formula. We refer to [Li24] to further details about SPX pricing together with equations, and to [MP18] for insights into turbocharging method.

3.3 SPX/VIX joint calibration

We now turn to the joint calibration of the quintic Ornstein-Uhlenbeck model to SPX European options, VIX European options, and VIX futures across several maturities simultaneously. As a practical matter, the SPX option maturities used in the calibration should extend about one month beyond the corresponding VIX maturities, since the VIX by construction encodes the market's expectation of volatility over the next 30 days. Fitting VIX futures alongside the VIX smile is not optional: futures prices enter directly into the construction of VIX implied volatilities, since these are obtained by inverting the Black-Scholes formula against the futures price of the underlying index rather than its spot price.

The model parameters are chosen to minimize a weighted combination of root-mean-squared errors (RMSE) across the three instrument classes: SPX implied volatilities, VIX implied volatilities, and VIX futures prices.

Concretely, we solve

$$\min_{\Theta} \Bigg\{ c_1 \sqrt{\textstyle\sum_{i,j} \left(\sigma^{\Theta}_{\text{spx}}(T_i,K_j) - \sigma^{\text{mkt}}_{\text{spx}}(T_i,K_j)\right)^2} + c_2 \sqrt{\textstyle\sum_{i,j} \left(\sigma^{\Theta}_{\text{vix}}(T_i,K_j) - \sigma^{\text{mkt}}_{\text{vix}}(T_i,K_j)\right)^2} + c_3 \sqrt{\textstyle\sum_{i} \left(F^{\Theta}_{\text{vix}}(T_i) - F^{\text{mkt}}_{\text{vix}}(T_i)\right)^2} \Bigg\}, \tag{49}$$

where, for a given maturity T_i and strike K_j, $\sigma^{\text{mkt}}_{\text{spx}}$ and $\sigma^{\text{mkt}}_{\text{vix}}$ denote respectively market-observed SPX and VIX implied volatilities, and $F^{\text{mkt}}_{\text{vix}}(T_i)$ the market VIX futures price. The corresponding model-implied quantities $\sigma^{\Theta}_{\text{spx}}$, $\sigma^{\Theta}_{\text{vix}}$, and F^{Θ}_{vix} are obtained from the model under parameter set Θ. Each of the three RMSE terms is summed independently over its own grid of maturities and strikes, so that a mispricing in one instrument type does not directly average out against another; the terms are combined only through the weights c_1, c_2, c_3. We take $c_1 = 1$, $c_2 = 0.1$, and $c_3 = 0.5$ throughout, giving primary weight to matching the SPX smile while quite anchoring the VIX curve and its futures level. We calibrate against daily SPX/VIX joint implied volatility surface data obtained from the Cboe (`https://datashop.cboe.com/`).

3.3.1 Extracting the forward variance curve $\xi_0(\cdot)$ ([Source: [AIL23]].

Using the log-contract replication formula [CM01], the forward variance curve is recovered from consecutive SPX maturities T_i via

$$\int_{T_i}^{T_{i+1}} \xi_0(s)\, ds = 2\left(\int_0^{S_0} \frac{P_0(K,T_{i+1})}{K^2}\, dK + \int_{S_0}^{+\infty} \frac{C_0(K,T_{i+1})}{K^2}\, dK\right) - 2\left(\int_0^{S_0} \frac{P_0(K,T_i)}{K^2}\, dK + \int_{S_0}^{+\infty} \frac{C_0(K,T_i)}{K^2}\, dK\right), \tag{50}$$

where $C_0(K,T)$ and $P_0(K,T)$ are the market call and put prices at strike K and maturity T. Since out-of-the-money option prices are not always quoted at every strike, the market SPX implied volatility surface is first interpolated slice by slice (e.g. via SVI or SABR, after checking for static arbitrage), and the fitted surface is used to evaluate the integrals in (50).

The curve ξ_0 is then approximated by a cubic spline through nodes (t_i, x_i), with $t_i = (T_i + T_{i+1})/2$ and $x_i = \sqrt{\int_{T_i}^{T_{i+1}} \xi_0(s)\, ds}$; squaring the spline guarantees positivity of the resulting forward variance curve. (A piecewise-constant curve on $[T_i, T_{i+1})$ is an equally valid alternative.)

During calibration, the optimizer is allowed to adjust both the model parameters Θ and the spline node values x_i, so that the forward variance curve itself becomes part of the joint fit to the SPX and VIX surfaces rather than a fixed input.

On 23 October 2017, this procedure yields a joint fit with the following calibrated parameters: $\rho = -0.6843$, $H = -0.0358$, $(\alpha_0, \alpha_1, \alpha_3, \alpha_5) = (0.5907, 1, 0.2893, 0.0549)$, and an accompanying adjustment of the stripped forward variance curve to further tighten the fit across SPX and VIX slices simultaneously (see Figure 4). Results of the calibration are shown in Figure 5.

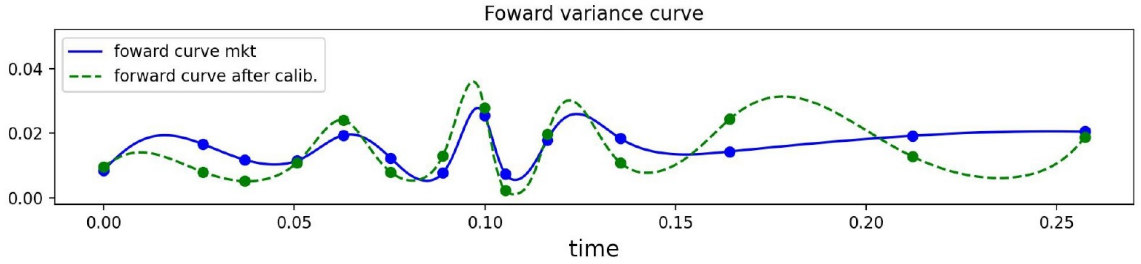


Figure 4: The blue line represent the forward variance curve stripped from market data using Carr-Madan log contract formula, the dotted green line is the adjusted forward variance curve as part of the calibration to jointly fit SPX/VIX smiles on 23 October 2017, with the round points representing cubic spline nodes. [Source: [AIL23]].

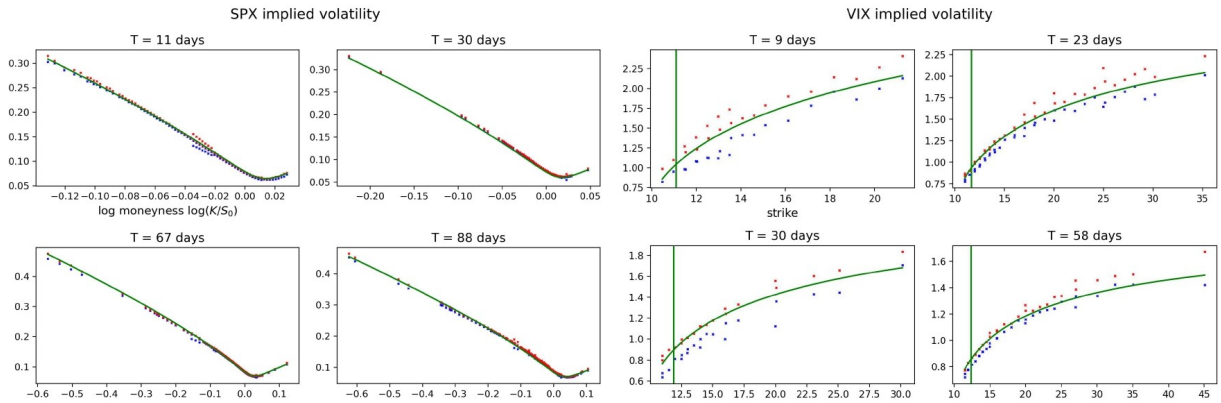


Figure 5: SPX–VIX smiles (bid/ask in blue/red) and VIX futures (vertical black lines) jointly calibrated with quintic OU volatility model (full green lines) for 23 October 2017. [Source: [AIL23]].

3.3.2 Using a parametric forward variance curves when calibrating fewer slices to boost efficiency

When calibrating among fewer slices, rather than stripping ξ_0 from market data, a parsimonious parametric form can be used instead:

$$\xi_0(t) = ae^{-bt} + c\left(1 - e^{-bt}\right), \tag{51}$$

where $a, b, c > 0$ are to be calibrated jointly with Θ. This form is considerably less flexible than a market-stripped curve, but it suffices when only a small number of slices are being fit, typically two SPX maturities together with one VIX maturity. Two such examples on 23 October 2017 are:

1. SPX options at 9 and 30 days together with VIX options at 9 days, with $\rho = -0.7316$, $H = -0.1382$, $(\alpha_0, \alpha_1, \alpha_3, \alpha_5) = (0.8169, 0.274, 0.1717, 0.0036)$, and curve parameters $a = 0.0084$, $b = 2.0436$, $c = 0.0441$;
2. SPX options at 53 and 88 days together with VIX options at 58 days, with $\rho = -0.7001$, $H = 0.141$, $(\alpha_0, \alpha_1, \alpha_3, \alpha_5) = (0.7558, 1, 0.0885, 0.4421)$, and curve parameters $a = 0.012$, $b = 2.027$, $c = 0.033$.

3.3.3 Time-dependent H for longer maturities

To extend the fit beyond maturities of three to four months, the coefficient H can be replaced by a time-dependent parametrization

$$H(t) = H_0 e^{-\kappa t} + H_\infty\left(1 - e^{-\kappa t}\right), \tag{52}$$

where $H_0, H_\infty, \kappa > 0$ are to be calibrated. Under this parametrization X remains a Gaussian Ornstein–Uhlenbeck process with time-dependent coefficients, and can still be simulated exactly; the expression for VIX_T^2 remains polynomial in X_T.

Combining (52) with a tweaked stripped forward variance curve and letting the mean-reversion speed ε float freely allows a joint fit of the SPX and VIX surfaces out to 18 months, using up to eight SPX slices and six VIX slices. We refer to [AIL23] for the corresponding calibration results on 23 October 2017.

3.3.4 Author's recommendations on which model's extension to use for different cases

Based on the above, a natural rule of thumb proposed in [AIL23] for choosing the forward curve and the parameters ε and H is:

- use the parametric curve (51) with fixed $\varepsilon = 1/52$ and constant H when fitting a single VIX slice together with two SPX slices;
- use a tweaked, market-stripped ξ_0 with fixed $\varepsilon = 1/52$ and constant H for joint fits spanning several maturities up to three or four months;
- use a tweaked, market-stripped ξ_0 with free ε and time-dependent H as in (52) for joint fits spanning several maturities up to 18 months.

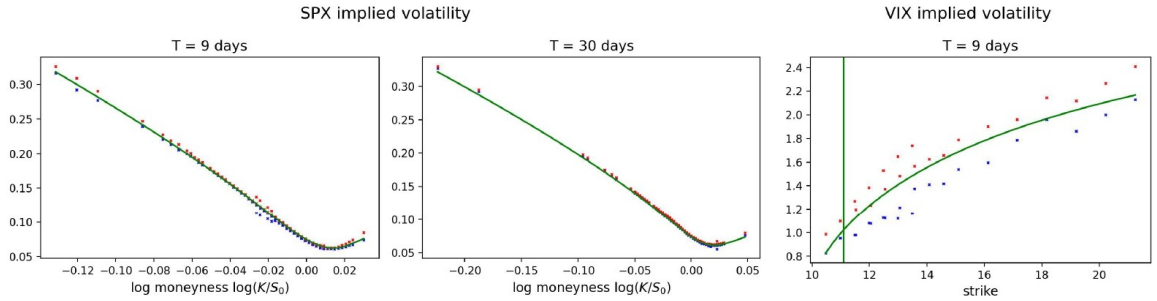


Figure 6: SPX–VIX smiles (bid/ask in blue/red) and VIX futures (vertical black lines) jointly calibrated with our model with the parametric forward variance curve for 23 October 2017. [Source: [AIL23]].

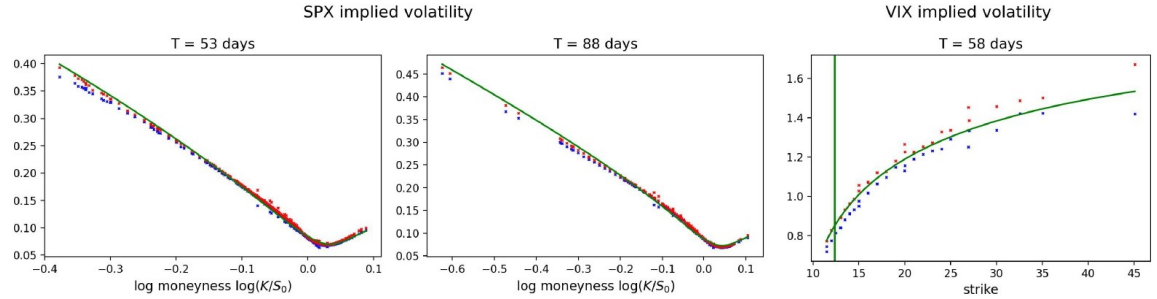


Figure 7: SPX–VIX smiles (bid/ask in blue/red) and VIX futures (vertical black lines) jointly calibrated with our model with the parametric forward variance curve for 23 October 2017. [Source: [AIL23]].

3.4 Model calibration errors

We show in Figure 8 the calibration error result by re-calibrating the model on a narrower, near-the-money strike range, then plotting each option's absolute pricing error as a multiple of half the bid-ask spread. A ratio below 1 means the model price falls inside the market's own bid-ask spread. The result is that the error ratio is mostly below 1 across both SPX and VIX slices, especially near the money, meaning the model's fit is generally as good as the market's own quoting precision allows.

Additional content on model calibration errors, in particular in the case of using a parametric forward variance curve, are given in [AIL23].

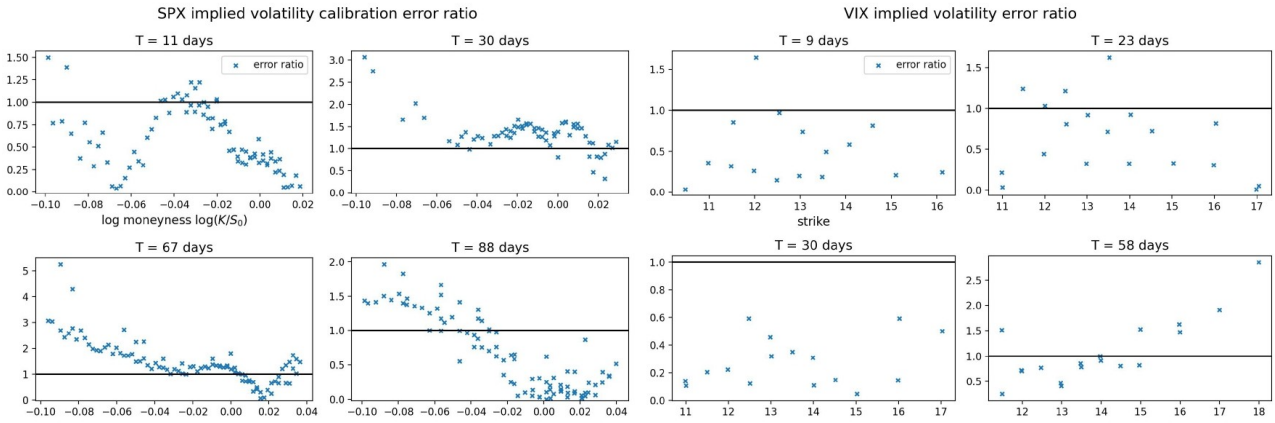


Figure 8: SPX–VIX implied volatility absolute calibration error as a multiplier of half of the bidask spread for 23 October 2017 using extracted forward variance curves. [Source: [AIL23]].

References

[AIL22] Eduardo Abi Jaber, Camille Illand, and Shaun Xiaoyuan Li. "Joint SPX–VIX calibration with gaussian polynomial volatility models: deep pricing with quantization hints". In: (2022). Available at SSRN: https://ssrn.com/abstract=4292544.

[AIL23] Eduardo Abi Jaber, Camille Illand, and Shaun Xiaoyuan Li. "The quintic Ornstein-Uhlenbeck volatility model that jointly calibrates SPX & Vix smiles". In: *Risk, Cutting edge section* (2023).

[Bos21] Sebastien Bossu. "Static replication of European options and dynamic replication of correlation swaps". ⟨NNT : 2021UPASI014⟩. 2021.

[Cbo25] Cboe. *Cboe Volatility Index. METHODOLOGY.* White paper, v20251022. Accessed: 22/08/2026. 2025.

[Cbo26a] Cboe. *Cboe VIX FAQ.* Accessed: 22/08/2026. 2026. URL: `https://www.tradingview.com/symbols/SPX/?exchange=SP`.

[Cbo26b] Cboe. *Volatility Index (VIX) Futures.* Accessed: 23/08/2026. 2026. URL: `https://www.cboe.com/tradable-products/vix/vix-futures/`.

[CM01] Peter Carr and Dilip Madan. "Towards a theory of volatility trading". In: *Option Pricing, Interest Rates and Risk Management, Handbooks in Mathematical Finance* 22.7 (2001), pp. 458–476.

[Dem+99] Kremisir Demeterfi et al. "More Than You Ever Wanted To Know * About Volatility Swaps _____ * But Less That Can Be Said". In: *QUANTITATIVE STRATEGIES RESEARCH NOTES Goldman Sachs* (1999).

[GJR20] Jim Gatheral, Paul Jusselin, and Mathieu Rosenbaum. "The quadratic rough Heston model and the joint S&P 500/VIX smile calibration problem". In: *Risk, Cutting edge section* (2020). ⟨hal-04573405⟩.

[Li24] Shaun Xiaoyuan Li. "The Quintic Volatility Model for SPX and VIX: A 360 Tour". ⟨tel-04910511v2⟩. Quantitative Finance [q-fin]. Université Paris 1 - Panthéon Sorbonne, 2024.

[MP18] Ryan McCrickerd and Mikko Pakkanen. "Turbocharging Monte Carlo pricing for the rough Bergomi model". In: *Quantitative Finance* 18.11 (2018), pp. 1877–1886.

[Neu94] Anthony Neuberger. "The Log Contract". In: *Journal of Portfolio Management* 20 (1994), pp. 74–80.

[Tra26a] TradingView. *S&P 500 Index.* Accessed: 22/08/2026. 2026. URL: `https://www.tradingview.com/symbols/SPX/?exchange=SP`.

[Tra26b] TradingView. *S&P 500 Index - Components.* Accessed: 22/08/2026. 2026. URL: `https://www.tradingview.com/symbols/SPX/components/`.